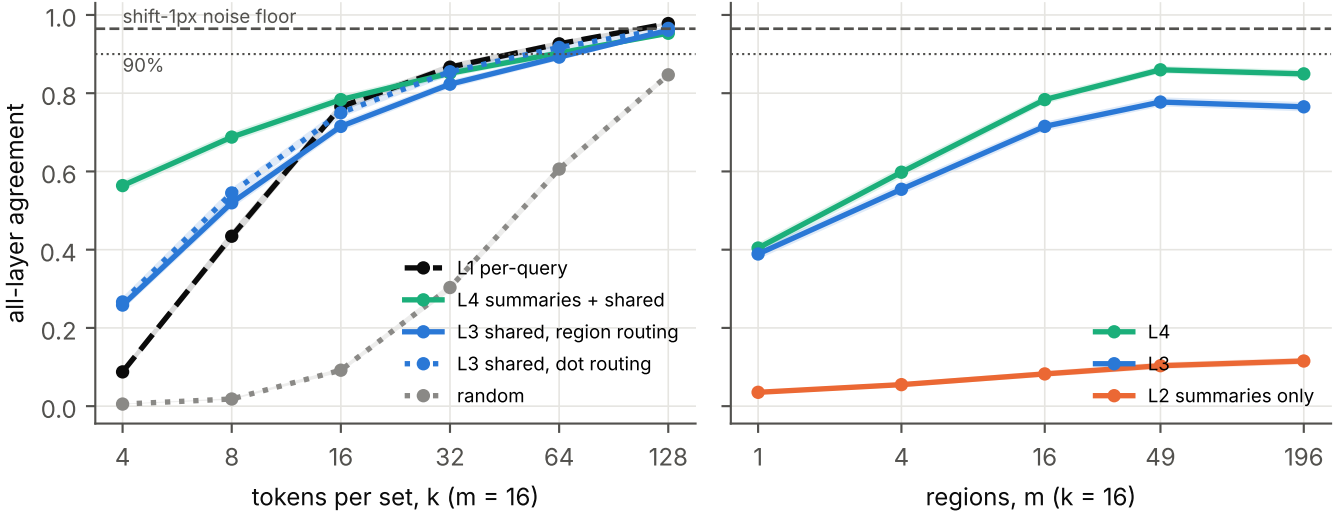All 12 layers replaced: sharing costs little against per-query top-k; summaries help only at small k